Time Series

2026/2027, Autumn Trimester

Course faculty	Ihor Miroshnychenko e-mail: imiroshnychenko@kse.org.ua
Department	Computer Science Department
Study program	Artificial Intelligence
ECTS credits	4 (120 hours)
Class hours	18 lecture hours and 22 practice hours (40 contact hours, 80 hours of self-study)
Course language	Materials in English; teaching in Ukrainian
Course format	Hybrid — every class can be joined online
Form of study	Full-time studies

Overview

This course comprehensively introduces time series analysis and forecasting using Python. Students will learn classical statistical methods (ETS, ARIMA, SARIMA, SARIMAX), machine learning approaches, and modern hybrid techniques for forecasting.

The course is built around applied work on real data. A lecture introduces a method; the practice session a week later works through it end to end on a shared example, arguing every decision out loud. The graded work — a lab on your own time series — is done afterwards, in your own time, and is due one to three days later. From Week 4 onward every lab uses the same individually assigned series, so by the end of the course you have built, compared and combined a whole family of models on one problem. The course culminates in a final project on a shared dataset.

Prerequisites

***Dependent courses*.**

- Must have: Probability Theory and Statistics
- Nice to have: Courses in data analysis or machine learning

Hard skills.

- Must have: Proficiency in Python (R is accepted in submissions, but templates and support are Python-only). Understanding of basic statistical concepts (mean, variance, correlation).
- Nice to have: Experience working with libraries for data analysis.

Soft skills. Problem-solving skills, analytical thinking, and the ability to work with data under time constraints.

Background and course rationale

Time series forecasting is a key data science skill used in finance, economics, energy, supply chain management, and many other domains. This course equips students with both theoretical understanding and practical skills to model and forecast time-dependent data.

This course differs from others in that:

- It uses Python, free, open source, and extremely powerful software.
- It includes dozens of real-world examples.
- It emphasizes graphical methods for exploring data, analyzing the reliability of fitted models, and presenting forecasting results.
- **It is practised under realistic constraints.** Forecasting in industry happens against a deadline and is judged against a benchmark. This course reproduces those conditions deliberately, and teaches that a more complex model is not automatically a better one.

This course addresses questions such as “How to analyze time series data?”, “What forecasting methods to use?” and “How to evaluate the accuracy of a forecast?”, providing students with practical skills in the Python environment.

Course aims

1. To familiarize students with the basic principles and methods of forecasting, and provide sufficient information for their effective use.

2. To teach students to apply various forecasting methods using Python tools.

3. To develop skills in analyzing time series data and evaluating forecast accuracy in the Python environment.

4. To prepare students to solve real-world forecasting problems in business and other fields using a practical, example-driven approach.

5. To inspire students to explore further more complex theoretical details and the latest developments in forecasting, including neural networks and foundation forecasting models.

Learning outcomes

By the end of the course, students will be able to:

1. Demonstrate a comprehensive understanding of the fundamental principles and methods of forecasting.
2. Apply various forecasting methods to time series data using Python libraries.
3. Analyze time series data using graphical methods for research, model validation, and presentation of results.
4. Specify a forecasting model manually from diagnostic evidence (ACF/PACF, differencing, seasonality) and justify that specification against what automatic model selection produces.
5. Evaluate the accuracy of forecasts against a benchmark, quantify forecast uncertainty, and select the most appropriate model for a specific task.
6. Judge when additional model complexity is warranted and when it is not — including recognising cases where a simple benchmark cannot be beaten.
7. Apply modern techniques such as neural networks and foundation models to forecasting tasks.

Methods of Teaching and Assessment

The course includes:

- **Lectures.** Theoretical material and core methods, one 80-minute lecture per week (Weeks 1–9), slides on Moodle. **Attendance is recommended**; every lecture can be joined online.
- **Practice sessions.** A worked example of the previous week's method on a shared dataset, every decision argued out loud. Nothing is submitted and attendance is not marked; notebooks are on Moodle. The practice runs one week behind the lecture, so a missed lecture never costs you the next lab.
- **Labs.** Seven graded labs (Weeks 2–4 and 6–9), done in your own time after the practice session and due one to three days later. From Week 4 each lab is a forecasting challenge on your own series.
- **Mid-term quiz.** A single proctored quiz taken electronically on Moodle.
- **Final project.** A two-session individual project on a shared dataset, published on GitHub and defended.

Course Faculty



Ihor Miroshnychenko, Ph.D. in Economics

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💼 Career Highlights:

- Teacher, Faculty of Computer Science, KSE
- Associate Professor, Department of Technology Management, FIT, KNU
- Lecturer, MBA, International Institute of Business

🔎 Research Interests:

- Data Science
- Statistics
- Data Analysis
- Machine Learning

💭 Teaching Philosophy:

I believe that data science is a powerful tool for understanding the world around us. My goal is to help students develop a deep appreciation for the role of data in decision-making and to equip them with the skills to analyze data effectively and communicate their findings clearly.

📋 Personal Interests:

- 🎸 Guitar playing
- 🚂 Traveling
- 📷 Photography
- 🎮 Gaming
- 📚 Reading
- ⚽ Football

✅ Interesting Facts:

- 🗻 Big fan of the Middle-earth universe.
- 🪶 Engaged in historical reconstruction
- 🤘 Metalhead

Course Structure

- ***Duration:*** *4 ECTS credits (120 hours), Autumn Trimester 2026/2027.*
- ***Weeks 1–9:*** *one lecture (80 min) and one practice session (80 min) per week, back to back. Every class can be joined online.*
- ***Week 10 (2–6 November):*** *two final-project sessions (80 min each), on different days.*
- ***Total:*** *9 lectures (18 hours), 11 practice sessions (22 hours), 80 hours of self-study.*
- **Week 1 is an onboarding session.** No coursework is assessed: the practice slot is used to set up environments, issue Track A datasets, and rehearse the submission mechanics end to end.

Grading Policy

General Grading Scheme

The course grade is the sum of all activities in the table below, capped at 100 points.

Task	***Count***	***Points per item***	***Total Points***
Weekly assignments	*7*	*8*	*56*
Mid-term quiz	*1*	*12*	*12*
Final project (challenge + report + defence)	*1*	*32*	*32*
Total			***100***
Activity bonus (extra)	*1*	*5*	*5*

Weekly labs

There are seven labs, each worth up to 8 points (56 points total). Each lab is published with its practice session and is due one to three days later; the exact deadline is announced with the lab and set according to the security and power situation that week. Work is submitted to Moodle and marked with written comments afterwards.

Assignments run in two tracks.

Track A — Foundational labs (Weeks 2–3)

The Track A labs use named, domain-rich datasets drawn from fpp3 / tsibbledata (e.g. aus_retail, vic_elec, pedestrian, PBS). The domain is visible and the history has meaning, so the work is about reading the data, not about scoring it. These labs are not competitive: at this point in the course there is no forecasting method or accuracy metric to compete on.

Criteria	**Points**
Data preparation	1
EDA and visualisation	4
Implementation	2

Code quality and reproducibility	1
Total	**8**

Track B — Forecasting challenges on your own series (Weeks 4, 6, 7, 8, 9)

The five challenge labs (Weeks 4, 6, 7, 8, 9) all use one series per student, issued in Week 4 and drawn deterministically from a curated pool by a hash of the student identifier. The series come from public economic statistics and are disguised on issue: identifiers removed, values rescaled, calendar shifted, history trimmed. Several series in the pool may be cut from the same source, so every model must be trained on your own series only. In the last lab (Week 9) you choose your route: combine your models, train a neural network on your series, use a foundation model zero-shot, or your own way.

Because every student works on a different series, discussing methods with your peers cannot become copying a result. Such discussion is encouraged and counts toward the activity bonus.

Criteria	**Points**
Forecast accuracy — beating the benchmark	3
Justification block	3
EDA and visualisation (in the last lab: your earlier models, rebuilt)	1
Code quality and reproducibility	1
Total	**8**

How the benchmark works.

- Accuracy is measured by a scale-free skill score: your forecast's MASE divided by your benchmark's MASE on the hidden horizon. This makes results comparable across different series.
- The **benchmark** is the best of the simple methods — naive, seasonal naive, drift, and mean — selected on a validation window. It is not fixed in advance to seasonal naive, because seasonal naive is trivially weak on some series and unbeatable on others.
- The pool of series is pre-screened: a series is only issued if it is seasonal and an automatic ETS and an automatic ARIMA each beat its benchmark. Every series you may be assigned is one that the course's methods are known to beat.
- Accuracy points are awarded against **your own series' benchmark**, not against your classmates: beating the benchmark scores 3, coming within 10% of it scores 2, and a worse result scores 1 with a required written explanation. Ranking against the class occurs only in the final project, where everyone forecasts the same dataset.
- In Weeks 4 and 7 accuracy is measured and reported but not marked: the method of those weeks cannot be guaranteed to beat the benchmark. Those labs are marked 1 / 3 / 3 / 1 — EDA, model and forecast, justification, code.

Deliverables for each challenge lab: a submission CSV of point forecasts (and prediction intervals where the task requires them), your code, and the justification block.

The justification block consists of three short fixed questions, specific to the week's topic. From Week 6 onward the task is structured so that you must specify the model manually first and justify that specification, and only then compare it with automatic model selection and explain any divergence. Automatic selection cannot be your first step. It is marked on the reasoning, not on the result: a model that lost to its benchmark earns full marks if the numbers are read correctly and the conclusion follows

from them. A justified step beyond the template — with a purpose, a sound method and an honest reading — earns up to +1 within the same lab, never above the lab's maximum.

Feedback. Marks and written comments are returned on Moodle before the next practice session, so there is always time to act on them.

Mid-term quiz

A single proctored quiz, held in the Week 5 practice slot and taken electronically on Moodle. It covers Chapters 1–3, 5 and 8 — the material completed before the quiz. Worth 12 points. A student who misses the quiz, for any reason and without documentation, takes it at an additional session or in an online oral interview on the same material, with immediate feedback.

Final project

The final project is an individual assessment in which students act as data scientists for an energy company. All students work with the same dataset, issued at the start of Session 1. Students do not select their own dataset.

The project runs across two sessions in Week 10 (2–6 November), held on different days. Both sessions are hybrid: you may work on campus or online.

Session 1 — Prediction challenge. Within the class period, produce point forecasts and 95% prediction intervals for a hidden 48-hour horizon and submit them before the end of the class. Submissions are scored on accuracy (RMSE, MAPE) and on the intervals (Winkler score), against a seasonal naive benchmark.

Between sessions. Improve your forecast at home: any model, any library. Using LLMs is allowed; describe how you used them in the repository's README.

Session 2 — Defence. Present your approach and results and answer questions, including two or three about the decisions in your own labs. This session is a public exchange: you will see how others attacked the same dataset, which methods worked, and which did not.

Deliverables: the Session 1 submission, the improved submission, a public GitHub repository with reproducible code and a report (Jupyter Notebook or PDF), and presentation slides.

Criteria and point breakdown (total: 32 points)

Criteria	Points	Stage
Valid Session 1 submission — correct format, submitted before the end of the class	5	Session 1
Beating the benchmark in Session 1	8	Session 1
Final skill score (class ranking)	5	Between sessions

Methodology and diagnostics — the choice of analyses is yours	8	Report
Code quality and reproducibility — must run without errors	2	Report
Defence and presentation	4	Session 2
Total	32	

On the final skill score. These 5 points are awarded on your absolute final accuracy on the shared dataset, ranked against the class — not on how much you improved relative to your own in-class result. Two gates apply: an in-class submission must have been made, and your final result must be no worse than it. Scoring improvement rather than absolute accuracy would reward deliberately underperforming in class, which is not what this course is for.

Activity bonus

Activity points reward contributing what you have learned, not being present. Points are awarded for substantive contributions:

- explaining your approach to the group;
- helping a peer reason about their own assigned series;
- asking substantive questions and offering insights during lectures, practice sessions and defences, including questions to other speakers;
- a correction to course materials.

Contributions made online during a class count exactly as those made in the room. Being present, and the number of remarks, earn nothing by themselves. Bonus points are added on top of your final grade, subject to the 100-point cap.

Missed work and interruptions

Attendance is not marked and no documentation is ever required. The following provisions exist so that circumstances outside your control do not decide your grade.

Situation	Provision
Lab deadline missed	If an air raid, an outage, illness or any other circumstance prevents you from meeting a lab deadline, tell the instructor within 48 hours after it and agree a new one: no explanation, no documentation, no cap on marks.
Missed mid-term quiz	Taken at an additional session or in an online oral interview on the same material, with immediate feedback.

Interruption during Final project, Session 1	If an air raid or a power or connection outage stops you during the session, your deadline is extended by the time lost; tell the instructor the same day. If the whole group is interrupted for more than 20 minutes, the session is rescheduled for everyone. The hidden horizon and the leaderboard stay unpublished until every Session 1 submission is in.
Unable to present at Session 2	Present online at an additional slot within a week; the questions and the criteria are the same.

Self-study (80 hours)

Self-study is a required component of the course, not optional reading. Labs are done in your own time, and each practice session assumes you have read the chapter it works through.

Activity	Hours
Required pre-reading and labs (Weeks 1–9)	35
Final project — improvement between the Week 10 sessions	20
Final project — report and presentation slides	15
Mid-term quiz preparation	10
Total	**80**

Academic integrity warning

Academic integrity is highly valued by the KSE community and its vast majority of students. We do not tolerate plagiarism and cheating. Very competitive essence of the program calls for absolutely equal treatment of all students. Even a single case of violation of academic integrity is a serious misdemeanor which may lead to unjust redistribution of grades and, consequently, overall rankings, possible grants, fee reductions and other merit-based awards.

The minimum penalty for a student caught cheating or plagiarizing during any practical task, a presentation or group and final projects will be a zero for this assignment with the consecutive decrease of the final course grade by at least one letter-grade. Repetitive violation of academic integrity may lead to the dismissal of the student from KSE. By enrolling in this course, students affirm that they are familiar with the KSE Code of Academic Integrity.

AI Tools usage policy

Scope. You are encouraged to explore and use generative AI as a learning tool, similar to how you would use a library, a search engine, or a discussion with a peer. You are ultimately responsible for the content and originality of all submitted work.

AI for coding assistance — permitted. You may use AI tools for coding: generating boilerplate, debugging, understanding concepts, or exploring alternative implementations. This includes all lab work.

Citation.

- ***Labs:*** *prompt-level citation is not required. The justification block replaces it: for every lab you must explain, in your own words, what you did and why. It is worth 3 of the 8 points, it is marked by hand, and you may be asked about it at the final-project defence.*
- ***Final project:*** *LLMs may be used for code and analysis; describe how you used them in the repository's README.*

AI for narrative in labs — not permitted. You may not use generative AI to write the narrative portions of your lab work: written analyses, explanations of your results, the justification blocks, or any text meant to represent your own understanding and interpretation.

Penalties. Any violation of this policy is treated as a breach of academic integrity, equivalent to plagiarism. Consequences may include a grade of zero on the assignment, failure of the course, and referral to the Academic Integrity Committee.

By submitting work in this course, you affirm that you have complied with this policy. If you have any questions about what constitutes appropriate use of AI, please ask the instructor before submitting your work.

Course Plan

Learning materials

Books:

1. Hyndman, R.J., & Athanasopoulos, G. (2021). Forecasting: principles and practice, 3rd edition, OTexts: Melbourne, Australia. OTexts.com/fpp3.
2. Hyndman, R.J., Athanasopoulos, G., Garza, A., Challu, C., Mergenthaler, M., & Olivares, K.G. (2024). Forecasting: Principles and Practice, the Pythonic Way. OTexts: Melbourne, Australia. Available at: OTexts.com/fpppy

Schedule

Week	Topics	Learning Materials	Learning Outcomes
1	**Introduction to forecasting + Time series graphics.**		
	• Course overview; • basic concepts of forecasting; • the forecaster's mindset; • time plots, seasonal and subseries plots, lag plots; • autocorrelation and white noise	Forecasting: principles and practice book (Ch. 1–2)	• Demonstrate a comprehensive understanding of the fundamental principles and methods of forecasting. • Analyze time series data using graphical methods for research, model validation, and presentation of results.
2	**Time series decomposition.**		

	● Transformations and adjustments; time series components; ● moving averages; ● classical decomposition; ● STL decomposition.	Forecasting: principles and practice book (Chapter 3)	● Demonstrate a comprehensive understanding of the fundamental principles and methods of forecasting. ● Analyze time series data using graphical methods for research, model validation, and presentation of results.
3	**The forecaster's toolbox.**		
	● Forecasting workflow; ● simple forecasting methods; ● fitted values and residuals; ● residual diagnostics; ● distributional forecasts and prediction intervals; ● forecasting with transformations and with decomposition; ● evaluating point forecast accuracy; ● time series cross-validation.	Forecasting: principles and practice book (Chapter 5)	● Demonstrate a comprehensive understanding of the fundamental principles and methods of forecasting. ● Analyze time series data using graphical methods for research, model validation, and presentation of results.
4	**Exponential smoothing.**		
	● Simple exponential smoothing; ● methods with trend; ● methods with seasonality;	Forecasting: principles and practice book (Chapter 8)	● Apply various forecasting methods to time series data using Python libraries.

	● a taxonomy of exponential smoothing methods; ● ETS models; ● estimation and model selection; ● forecasting with ETS.		● Evaluate the accuracy of forecasts against a benchmark, quantify forecast uncertainty, and select the most appropriate model for a specific task.
5	**Time series regression models (bonus topic) | Mid-term Quiz**		
	● Linear models for time series; ● trend and seasonal dummies; ● selecting predictors; ● forecasting with regression	Forecasting: principles and practice book (Chapter 7)	● Apply various forecasting methods to time series data using Python libraries. ● Evaluate the accuracy of forecasts against a benchmark, quantify forecast uncertainty, and select the most appropriate model for a specific task.
6	**ARIMA models, part 1.**		
	● Stationarity and differencing; ● autoregressive models; ● moving average models; ● non-seasonal ARIMA; ● **manual order identification from ACF/PACF**; ● estimation.	Forecasting: principles and practice book (Chapter 9)	● Apply various forecasting methods to time series data using Python libraries. ● Specify a forecasting model manually from diagnostic evidence (ACF/PACF, differencing, seasonality) and justify that specification against what automatic model selection produces.

			• Evaluate the accuracy of forecasts against a benchmark, quantify forecast uncertainty, and select the most appropriate model for a specific task.
7	**ARIMA models, part 2.**		
	• Seasonal ARIMA; • automatic order selection; • ARIMA vs ETS; • **SARIMAX / dynamic regression models**	Forecasting: principles and practice book (Chapter 9-10)	• Apply various forecasting methods to time series data using Python libraries. • Specify a forecasting model manually from diagnostic evidence (ACF/PACF, differencing, seasonality) and justify that specification against what automatic model selection produces. • Evaluate the accuracy of forecasts against a benchmark, quantify forecast uncertainty, and select the most appropriate model for a specific task.
8	**Advanced forecasting methods.**		
	• Complex seasonality; • Prophet; • vector autoregressions; • bootstrapping and bagging.	Forecasting: principles and practice book (Chapter 12)	• Judge when additional model complexity is warranted and when it is not — including recognising cases where a simple benchmark cannot be beaten.

			• Apply modern techniques such as neural networks and foundation models to forecasting tasks.
9	**Neural network forecasting.**		
	• Neural forecasting; multilayer perceptron; • modern architectures; • optimization objectives; • exogenous variables; • hyperparameter optimization; • transfer learning; • foundation models: zero-shot forecasting, pretraining corpora and leakage (practice session).	Forecasting: principles and practice book (Chapter 14-15)	• Judge when additional model complexity is warranted and when it is not — including recognising cases where a simple benchmark cannot be beaten. • Apply modern techniques such as neural networks and foundation models to forecasting tasks.
10	**Final project (2–6 November)**		
	• Session 1 (hybrid): prediction challenge on a shared dataset, 48-hour hidden horizon. • Session 2 (hybrid, different day): defence and exchange of approaches.		• Apply various forecasting methods to time series data using Python libraries. • Evaluate the accuracy of forecasts against a benchmark, quantify forecast uncertainty, and select the most appropriate model for a specific task.

			• Judge when additional model complexity is warranted and when it is not — including recognising cases where a simple benchmark cannot be beaten. • Apply modern techniques such as neural networks and foundation models to forecasting tasks.